Topology and Manifolds

MH3600

INTRODUCTION

Assoc. Prof. François J. M. Gay-Balmaz

Division of Mathematical Sciences
School of Physical and Mathematical Sciences
Nanyang Technological University

Lectures and Tutorials

- Monday (Lectures) 9:30-11:20. Venue SPMS-LT4
- Wednesday (Lecture + Tutorial) 10:30-12:20. Venue SPMS-LT4
- Slides and tutorials will be made available in the corresponding sections in NTU Learn.

Brief course description

- This course is a first introduction to topology and calculus on manifolds: the most general framework to discuss the notions of:
  – convergence, continuity (topology)
  – differentiability (calculus)
- Natural generalization to infinite-dimensional and non-Euclidean spaces of the ideas that you learnt for $\mathbb{R}^n$ in Calculus I, II, and III.
- These tools are fundamental for other fields of mathematics: algebraic topology, functional analysis, differential / Riemannian / symplectic / geometry, Lie theory, machine learning, numerical analysis, ...
- They also have strong ties with important applications in the physical sciences and engineering like dynamical systems, mechanics, fluid dynamics, mathematical physics, control theory, ...
- The aim of this course is to enable you to formulate and solve mathematical problems using the ideas and the formalism coming from topology and global analysis.

Main contents

1. Sets, functions, relations, finite and infinite sets, cardinalities.
2. Topological spaces. Basis of a topology. Subspace, product, box, and quotient topologies.
3. Continuous functions. Homeomorphisms, embeddings.
4. Topologies induced by a metric, by a norm, or by an inner product. Function spaces and uniform convergence.
5. Completeness and Fixed Point Theorems.
6. Connectedness and compactness.
7. Manifolds. Charts and atlas. Atlas equivalence and manifold topologies. Examples. Submanifolds, products, and mappings.
8. The tangent and the cotangent bundle. Subimmersions, immersions, and transversality.
9. Vector fields, flows, and dynamical systems.

References

Main sources are:

- Munkres, James R. Topology, a First Course. Second edition. Pearson, 2014. ISBN: 978-1292023625.
- Willard. S. General Topology. Dover. 1994. ISBN: 978-0486434797.
- Abraham, R., Marsden, J. E., and Ratiu, T. S. Manifolds, Tensor Analysis, and Applications. Second Edition. 1988. ISBN: 978-1461269908

But will also be using:

- Simmons, G. F. Topology and Modern Analysis. McGraw-Hill. 1963.

Countless ressources for online problems. The solutions of all problems in Munkres book can be found here.

Some advice

- The material delivered during the lectures does not suffice. Independent reading is required.
- Try to keep up with the material as we deliver it in the lectures. Otherwise lectures will be useless.
- Work out problems by yourself beyond those covered in the tutorials. This is a "conceptual" not a "recipe" course.
- Looking at solved problems has limited value in this class. This is not calculus anymore!
- One important objective in the tutorial session is learning to correctly present and discuss mathematical results. We will learn to write and talk mathematics.
- If you find typos, errors, or unclear statements, let me know!

Assessment

Your grade on the course will be determined by:

- Assignment, 20% (approximately week #6, submission on week #7)
- Tutorial presentations, 10%. Tutorial sessions will be conducted by students. Credit will be given for the solution of the problem and its presentation. Students will be designed one week in advance and can prepare the solution at home to be presented in class during the next tutorial.
- Project presentation, 10%. To be conducted individually during the last week of the course. The topic will be chosen by the student and will be validated by the instructor. Topic has to be chosen by week #9. Further instructions will follow.
- Final exam, 60%

Important Note

- – A late submission for an assignment is not accepted.
  – No make-up assignments or (tutorial or project) presentations will be arranged.
  – A student who is late for an assignment or (tutorial or project) presentation without valid Leave of Absence will be given zero mark.
  – In case of a valid reason for absence, the total course marks would subsequently be rescaled to a base of 100%.
- The required number of required tutorial presentations and the duration of the project presentation will be adjusted as a function of the total number of students attending the class.

Your Instructor

- Name: François Gay-Balmaz.
- Associate professor, Mathematical Sciences, School of Physical and Mathematical Sciences (SPMS).
- PhD at EPFL (Switzerland), Postdoc at Caltech (USA).
- 2010-2023: CNRS, Ecole Normale Supérieure in Paris.
- Email: francois.gb@ntu.edu.sg. Office: SPMS-MAS-05-41.
- Research: at the intersection of Geometry, Numerics, Fluid Dynamics, Physics.

  - Geometric methods in fluid dynamics (atmosphere & ocean dynamics for weather, plasma physics for fusion, ...)
  - Structure preserving discretization (numerical methods)
  - Variational principles in nonequilibrium thermodynamics
  - Geometry and modeling of quantum-classical systems

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